

5.3.11

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Question

Find whether the following pair of linear equations are consistent or inconsistent:

$$5x - 3y = 11, \quad -10x + 6y = 22 \quad (1)$$

.

Theoretical Solution

Let us solve the given equation theoretically and then verify the solution computationally

According to the question,

Given two lines

$$\mathbf{A} = \begin{pmatrix} 5 & -3 \\ -10 & 6 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 11 \\ 22 \end{pmatrix} \quad (2)$$

(3)

$$\mathbf{AX} = \mathbf{B} \quad (4)$$

rank of $\mathbf{A} = 1$

rank of $\begin{pmatrix} \mathbf{A} & \mathbf{b} \end{pmatrix} = 2$

As both ranks are not equal the pair of linear equations are inconsistent

```
#include <stdio.h>

int main() {
    // Coefficients of equations
    //  $5x - 3y = 11 \rightarrow a1=5, b1=-3, c1=11$ 
    //  $-10x + 6y = 22 \rightarrow a2=-10, b2=6, c2=22$ 
    float a1 = 5, b1 = -3, c1 = 11;
    float a2 = -10, b2 = 6, c2 = 22;

    // Ratios
    float ratio1 = a1 / a2;
    float ratio2 = b1 / b2;
    float ratio3 = c1 / c2;

    if (ratio1 == ratio2 && ratio2 == ratio3) {
        printf("The equations are consistent and have infinitely  
many solutions.\n");
    }
}
```

```
else if (ratio1 == ratio2 && ratio2 != ratio3) {  
    printf("The equations are inconsistent (no solution).\n")  
    ;  
}  
else {  
    printf("The equations are consistent and have a unique  
        solution.\n");  
}  
  
return 0;  
}
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Define x values
x = np.linspace(-10, 10, 400)

# First equation:  $5x - 3y = 11 \rightarrow y = (5x - 11)/3$ 
y1 = (5*x - 11)/3

# Second equation:  $-10x + 6y = 22 \rightarrow y = (10x + 22)/6$ 
y2 = (10*x + 22)/6

# Plot the lines
plt.plot(x, y1, label="5x - 3y = 11")
plt.plot(x, y2, label="-10x + 6y = 22")
```

Python Code

```
# Labels and title
plt.xlabel("x-axis")
plt.ylabel("y-axis")
plt.title("Graph of Linear Equations")
plt.axhline(0, color='black', linewidth=0.5)
plt.axvline(0, color='black', linewidth=0.5)
plt.grid(True, linestyle="--", alpha=0.6)

# Show legend
plt.legend()

# Save the figure
plt.savefig("lines.png", dpi=300)

# Show the plot
plt.show()
```

